Efficient Bayesian Inference for Multivariate Factor Stochastic Volatility Models

Kastner et al.2017)

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Outline

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From Univariate to Multivariate Stochastic Volatility

- **Recall Assignment Task 2:** We modelled daily CHF/USD returns with a t_ν-distributed error and latent weights ω_i, giving each observation its own effective variance σ^2/ω_i — a *univariate* stochastic volatility model.
- **Natural extension:** With m assets we need not only time-varying variances but also time-varying *covariances*. The full covariance matrix has $m(m+1)/2$ free elements — growing **quadratically** in m.
- **The curse of dimensionality:** For $m = 26$ exchange rates this gives 351 covariance parameters *per time point*. Direct modelling becomes infeasible.
- **Solution:** Impose a low-dimensional **factor structure** that reduces the parameter count while retaining flexibility for volatility clustering and co-movement.

The Factor SV Model: Specification

For $t = 1, \ldots, T$ and m observed returns $\mathbf{y}_t$ driven by $r \ll m$ latent factors $\mathbf{f}_t$:

Observation & Factor Equations

$$\mathbf{y}_t = \boldsymbol{\Lambda}\mathbf{f}_t + \varepsilon_t, \qquad \mathbf{f}_t \mid \mathbf{h}_t \sim \mathcal{N}_r\big(\mathbf{0},\ V_t(\mathbf{h}_t^V)\big)$$

with $\varepsilon_t \mid \mathbf{h}_t \sim \mathcal{N}_m\big(\mathbf{0}, U_t(\mathbf{h}_t^U)\big)$

- $\boldsymbol{\Lambda}$: unknown $m \times r$ **factor loadings** matrix
- $U_t = \mathrm{diag}\big(e^{h_{1t}}, \ldots, e^{h_{mt}}\big)$: **idiosyncratic** (series-specific) variances
- $V_t = \mathrm{diag}\big(e^{h_{m+1,t}}, \ldots, e^{h_{m+r,t}}\big)$: **factor** variances
- Each log-variance follows an independent **AR(1)** process:
  $h_{it} = \mu_i + \phi_i(h_{i,t-1} - \mu_i) + \sigma_i \eta_{it}$

The time-varying conditional covariance matrix is:

$$\mathrm{cov}(\mathbf{y}_t \mid \mathbf{h}_t) = \boldsymbol{\Lambda} V_t \boldsymbol{\Lambda}' + U_t$$

Identification

- **Scaling ambiguity:** Multiplying column j of $\mathbf{\Lambda}$ by c and dividing f_{jt} by c leaves $\mathbf{y}_t$ unchanged — the model is not identified without additional constraints.
- **Classical fix:** Set $\lambda_{jj} = 1$ and treat the factor variance level as free. This imposes a series ordering and creates "leading" factors.
- **Paper's approach:** Fix the level $\mu_{m+j} = 0$ for each factor log-variance (i.e. factor variance averages to 1), while leaving λ_{jj} *unrestricted*.
- **Sign switching:** Each column of $\mathbf{\Lambda}$ is identified only up to a sign flip — resolved in post-processing.

Why it matters

Unresolved non-identifiability causes MCMC chains to be multimodal and essentially useless. The paper deliberately *exploits* the non-identifiability as a computational tool (see ASIS, Section 4).

Bayesian Inference: Gibbs Sampler

Data augmentation with latent volatilities $\mathbf{h}$ and factors $\mathbf{f}$ makes conditional posteriors tractable — exactly as in the t-model from Task 2.

Algorithm 1 – Standard Gibbs Sampler

(a) **Univariate SV updates** ($m + r$ independent problems): given $\mathbf{f}$ and $\mathbf{\Lambda}$, each series/factor reduces to a univariate SV model via the log-squared observation trick. Use the `stochvol` R package as a plug-in.

(b) **Sample loadings $\mathbf{\Lambda}$**: conditional on $\mathbf{f}$ and $\mathbf{h}$, each row of $\mathbf{\Lambda}$ is a standard normal regression posterior.

(c) **Sample factors $\mathbf{f}_t$**: conditional on $\mathbf{\Lambda}$ and $\mathbf{h}$, each $\mathbf{f}_t$ is a multivariate normal posterior.

Link to lecture (Week 2–3): Steps (b) and (c) are identical in structure to the conjugate Normal–Normal updates for μ and the ω_i in Task 2. Step (a) extends the MH step for ν.

Problem: Standard Gibbs Mixes Poorly

- The factor loadings $\mathbf{\Lambda}$ and the latent factors $\mathbf{f}_t$ are **strongly correlated** in their joint posterior. Updating them sequentially leaves the chain nearly stuck.
- In simulation studies with $m = 10$, $r = 2$, $T = 1000$: the standard sampler yields **inefficiency factors (IFs) of 5000+** for some datasets — meaning you need 5000 MCMC draws to equal one independent draw.
- Even after 5 million iterations the chain may not have converged.

Inefficiency Factor (IF)

Ratio of the MCMC numerical variance to the variance from i.i.d. draws. An IF of k means you need k times as many MCMC draws to achieve the same accuracy as independent sampling. Estimated via integrated autocorrelation time (`coda` package).

Ancillarity-Sufficiency Interweaving Strategy (ASIS)

Key idea: Two *mathematically equivalent* parameterisations are each efficient in different regions of the parameter space. Switching between them at every iteration combines the best of both worlds (Yu and Meng2011).

Shallow Interweaving

- Transform to alternative param. where $\lambda_{jj} = 1$
- Resample λ_{jj} from a **Generalised Inverse Gaussian** (GIG) posterior (conjugate)
- Transform back
- Affects $\mathbf{\Lambda}$ and $\mathbf{f}$ only
- $\approx$ 2–8$\times$ efficiency gain

Deep Interweaving

- Additionally shift log-volatility path: $h^*_{m+j,t} = h_{m+j,t} + 2\log|\lambda_{jj}|$
- Resample λ_{jj} via **independence MH** in this extended space
- Affects $\mathbf{\Lambda}$, $\mathbf{f}$ *and* $\mathbf{h}$
- Up to **400**$\times$ efficiency gain

No tuning required: The method is fully automatic — no user adjustment of proposal variances (contrast with the c_ν tuning in Task 2).

Simulation Evidence

Setup: $m = 10$ series, $r = 2$ factors, $T = 1000$ observations, 100 replications, 5.1 million draws per run.

	No interweaving	**Shallow**	**Deep**
λ_{11} (average IF)	> 1000	≈ 200	≈ 5
Median IF across $\mathbf{\Lambda}$	$\sim$400–5000	$\sim$50–400	$\sim$3–20
Efficiency gain vs. no interweaving	—	2–8$\times$	up to **400**$\times$
Factor $f_{1,T}$ (IF)	high	moderate	low
Factor log-var $h_{m+1,T}$ (IF)	high	high	substantially reduced

Source: Tables 1–2 of Kastner et al.2017). IFs estimated via `coda`.

Table 1: Average inefficiency factors for the factor loadings matrix.

- Deep interweaving trace plots show **immediate mixing**; the standard sampler barely moves even after millions of draws (Figure 1 of the paper).
- Log predictive Bayes factors (cf. Week 3) correctly identify $r = 2$ as the true number of factors.

Application: 26 EUR Exchange Rates (2005–2015)

- **Data:** $m = 26$ daily EUR exchange rates, $T = 2649$ log-returns (demeaned). Includes outlier episodes: CHF spike Jan. 2015, RUB collapse Dec. 2014.
- **Model choice:** $r = 4$ factors selected via log predictive Bayes factors (Week 3, slide 8). 500 000 post-burn-in draws.
- **Interpretable factors:**
  - **Factor 1:** USD/CNY-driven global dollar factor (USD, CNY, HKD load highly)
  - **Factor 2:** Eastern European / EM currencies (ZAR, HUF, PLN, TRY; JPY loads *negatively*)
  - **Factor 3:** Commodity currencies (AUD, NZD, CAD, ZAR)
  - **Factor 4:** Tiger Cub economies (MYR, KRW, PHP, SGD)
- **Time-varying correlations:** CNY and HKD nearly perfectly correlated with USD throughout; RUB–USD correlation fell from ≈ 0.9 to ≈ 0.4 during 2008–2009.
- **Robustness:** Idiosyncratic SV components absorb outliers (CHF, RUB) without distorting the factor structure — analogous to how ω_i absorbed outliers in Task 2.

Takeaways & Practical Relevance

Theoretical connections to the lecture:

- **Data augmentation (Week 2–3):** The factor SV Gibbs sampler is a direct generalisation of the ω_i-augmented sampler in Task 2. Both make conditionally intractable models tractable.
- **Model selection (Week 3):** Log predictive Bayes factors (Kass–Raftery scale) are used to select r — identical framework to the marginal likelihood comparisons in the lecture.
- **Shrinkage (Week 3):** The Gaussian prior on $\mathbf{\Lambda}$ acts as a Ridge penalty, regularising small loadings and preventing spurious factors.

Practical use:

- The `factorstochvol` R package implements the full algorithm.
- Deep interweaving adds negligible computational cost but yields orders-of-magnitude better mixing — always prefer it.
- Even a correctly specified Gibbs sampler can be **practically useless** without good mixing. ASIS is a general remedy applicable beyond this specific model.

Bibliography

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